1 A machine fills 1000 bottles in 5 hours.

Work out how many hours it would take the machine to fill 1200 bottles.

6 hours **(2 marks)**

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2 It costs £0.75 to buy 5 bananas.

Work out how much it would cost to buy 7 bananas. 1.05

(2 marks)

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3 3 tins of beans and 4 tins of tomatoes costs £2.73.

5 tins of beans costs £1.55. 29p

Work out how much one tin of tomatoes costs.

(2 marks)

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4 There are 500 sheets in a pack of paper. 500 sheets of paper weigh 2.5kg.

Work out the weight of 50 sheets of paper.

0.25kg **(2 marks)**

---

5 It takes 2 painters 4 days to complete a job.

Work out how many days it would take 1 painter to complete the same job.

(2 marks)

8 days

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6 It takes 3 machines 2 days to produce a batch of products

Work out how long it would take 1 machine to produce the same batch of products.

6 days **(2 marks)**

---

7 It takes 3 painters 6 days to complete a job.

Work out how many days it would take 2 painters to complete the same job.

9 days **(2 marks)**

---

8 It takes 5 machines 6 hours to produce 1000 DVDs

Work out how long it would take 4 machines to produce 1000 DVDs.

7.5 hours **(2 marks)**

---

9 x is inversely proportional to y.

x is given by the formula: $x = \frac{1000}{y}$

Find the value of x when $y = 50$ 20

(2 marks)

---

10 y is directly proportional to x.

y is given by the formula: $y = 0.4x$

Find the value of y when $x = 6$ 2.4

(2 marks)

11 The weight of a piece of wire (w grams) is directly proportional to its length (l cm).

w is given by the formula: $w = 30l$

Find the length of a wire weighing 75 grams.

2.5cm

(2 marks)

---

12 The force, F, between two magnets is inversely proportional to the square of the distance, x cm, between them.

F is given by the formula:

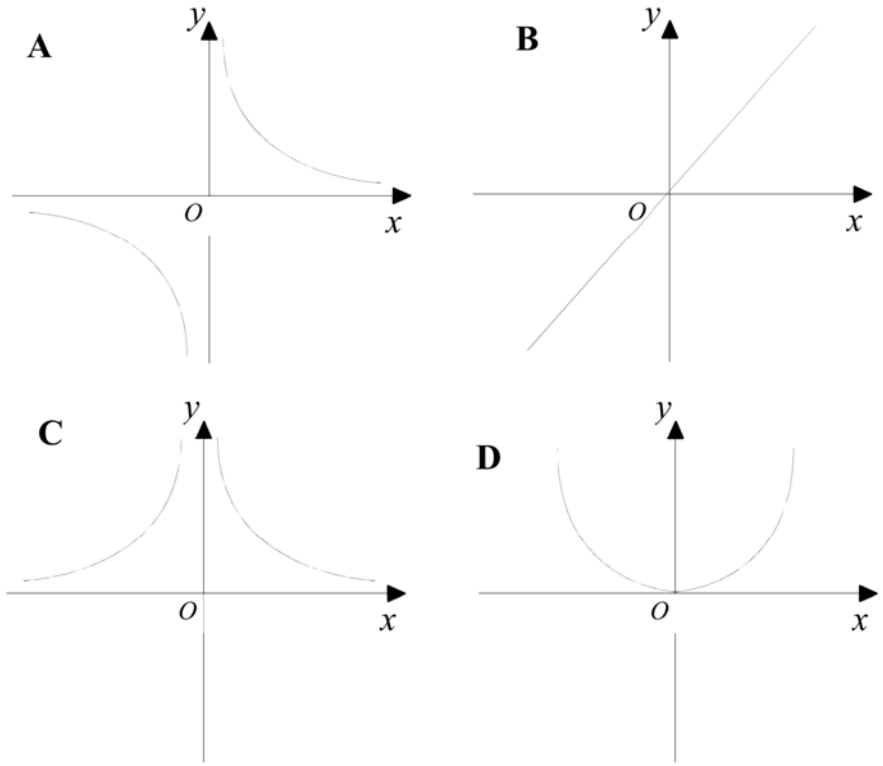
$$F = \frac{36}{x^2}$$

4

Find the Force when two magnets are 3 cm apart.

(2 marks)

13 Here are four graphs.



Sketch each graph and match with a statement in the table below.

Proportionality relationship	Graph letter
y is directly proportional to x	B
y is inversely proportional to x	A
y is directly proportional to x^2	D
y is inversely proportional to x^2	C

(2 marks)